Model performance comparison 2

Random Forest Experiment

Criterion	Estimator	R2 score	Status
Default	50	0.944633	Good
Squared_error	100	0.94043	Good
Absolute_error	50	0.941193	Good
Absolute_error	100	0.945909	Good
friedman_mse	50	0.938895	Low
Poisson	50	0.94635	Best
Poisson+max_depth=None	50	0.946354	Best
Poisson+monotonic_cst	50	0.94635	Good

Observation:

In Random Forest, poisson criterion gave the highest R2 score even without scaling. Increasing n_estimators from 50 to 100 improved absolute_error but decreased squared_error. friedman_mse gave the lowest score of 0.938895. Overall poisson performed best.

Conclusion:

In Random Forest, criterion=poisson, estimator=50 and max_depth=None is the overall best model with a score of 0.9463549. This outperforms Decision Tree (0.931135) and SVR Linear (0.895077).

Final Model: Random Forest (poisson + estimator=50) = 0.9463549